

A student team building an autonomous boat from the ground up.



HOW WE MODEL IT

First-principles mixed-integer linear programming

We build the model from physics up. Each constraint cuts the space of possible boats, and the optimizer finds the fastest one left. Hull resistance comes from **Savitsky's planing-hull model**. As we build and test, we keep updating the model and adding constraints: a hybrid of simulation and empirical data.

ENGINEERS FROM ACROSS UW-MADISON

Each segment of the boat is covered, and members have worked at Tesla, Northrop Grumman, Xcel Energy and Milwaukee Tool.

Industrial

Optimization, composites and manufacturing

Mechanical

Structures, CAD and fabrication

Electrical

Power, battery and high voltage

Computer

Embedded, firmware and software

OUR APPROACH

Hull

Plane well and stay light.

- Hybrid planing composite hull
- Shape tuned by optimization
- Resistance checked in CFD first

Electric

Every watt-hour to speed.

- High-voltage battery storage
- Propulsion sized by the model
- Propulsion and cooling co-designed

Autonomous

Find the way, unaided.

- GPS, IMU and LIDAR sensing
- Firmware protects the battery
- Software plans the route

TIMELINE

Oct 25
Hull design

Nov 15
Electrical design

Nov 25
Hull built

Jan 25
First drive

Feb 15
First fast drive

Mar 15
Tuning and testing